Rajan Patkar

rajpatkar.github.io/contact (815) 408-0845 rajan.patkar.email@gmail.com

Relevant Experience:

Research Assistant (prev. Intern) in Cybersecurity at the University of Illinois (May 2025 – now):

- Evaluated DPU performance in monitoring kernel health of network of smart devices
- Consulted on implementation of CIE practices to mitigate simulated OT network attacks
- Aiming to build on previous work to employ DPU as a secure root of trust for OT systems

Cybersecurity Intern at the University of Illinois Information Trust Institute (May – Aug. 2024):

- Crafted hardware/software suite to determine efficiency of large-scale SSL offloading
- Designed network agents in C to record packet timing data for network performance
- Utilized Python automation package to deploy network agents and run experiments
- Presented experiment findings to recommend OT networks continue with on-device SSL

Corporate Engineering Intern at Caterpillar (May – Aug. 2022):

- Automated collection and processing of machine data reports from database with Python
- Identified serial data communication problems (CAN + J1939) on engine control modules
- Developed field-ready solutions using electronics test benches and network diagnostics

STEM Student Trainee at Caterpillar Trimble Control Technologies (Jun. – Aug. 2021):

- Automated testing of calibration software for hydraulic excavators using MATLAB

Technology Development Volunteer Intern at The Farmlink Project (Jun. – Aug. 2020):

- Built automatic tax deduction tool for farmers' food donations with JavaScript & G Suite

Education:

M.S. in Electrical and Computer Engineering at University of Illinois Urbana-Champaign

- Goal is to complete research that enhances security of cyberphysical systems
- Continuing from B.S. in Computer Engineering at Illinois (grad. May 2025) – 3.72 GPA
- Focus on cybersecurity as a former SFS (Scholarship for Service) CyberCorps Scholar
- Study abroad semester (fall 2024) at Hanyang University in Seoul, SK

Skills:

- Proficient in C, C++, x86 assembly, Python, MATLAB, R, Java, ArcGIS, JS, and HTML
- Applied side-channel, SQL injection, XSS, CSRF, shellcode, and hash collision tactics
- Proficient interpersonal communicator (native English/Marathi, B2 Spanish, B1 Korean)

Projects:

- Piloted analyzing cellphone video to triangulate event location/timing using OSINT
- Constructed reliable serial communication protocol to transmit motor and sensor signals